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Course Title: Computer Networks

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Lab NO: 04

Department of Software Engineering

✓ **Lab Report**

Implementation and Evaluation of a Star Topology Using Cisco Packet Tracer

Overview

This laboratory task focused on the design and analysis of a Star Topology using Cisco Packet Tracer. A basic Local Area Network (LAN) was created using one Layer 2 switch and three personal computers. After configuration, connectivity was tested through ping commands, and packet transmission was examined in both Realtime and Simulation modes.

The activity provided a clear understanding of how data packets travel within a star topology and how networking devices operate at both the data link and network layers.

Introduction

A star topology is among the most commonly implemented network structures in modern LAN environments. In this setup, all devices such as computers, laptops, and servers are connected to a central device, typically a switch. This architecture offers advantages like easy network expansion, simplified troubleshooting, and better performance compared to other topologies like bus or ring.

Cisco Packet Tracer, developed by Cisco, is a widely used simulation tool that enables users to build, configure, and troubleshoot networks virtually. In this exercise, a simple star network was constructed, IP addresses were assigned, connectivity was verified using ping, and packet movement was analyzed through simulation mode.

1. Objectives

The goals of this experiment were:

- To explore the design and functionality of a star topology
- To create a small LAN using a switch and multiple PCs in Cisco Packet Tracer
- To assign appropriate IP addresses for communication
- To test connectivity using the ping command
- To monitor packet transmission using Simulation Mode
- To distinguish between control packets and user-generated data

2. Theoretical Background

In a star topology, all nodes are linked to a central switch. The switch learns the MAC addresses of connected devices and forwards data frames only to the correct destination port, minimizing unnecessary traffic (unlike hubs).

Key benefits include:

Simple addition or removal of devices

Isolation of faults (a single link failure does not disrupt the entire network)

Enhanced network performance and security

For communication to occur successfully, all devices must have IP addresses within the same subnet.

3. Tools and Equipment

Cisco Packet Tracer (latest version)

One 2960-24TT switch

Three PC-PT end devices

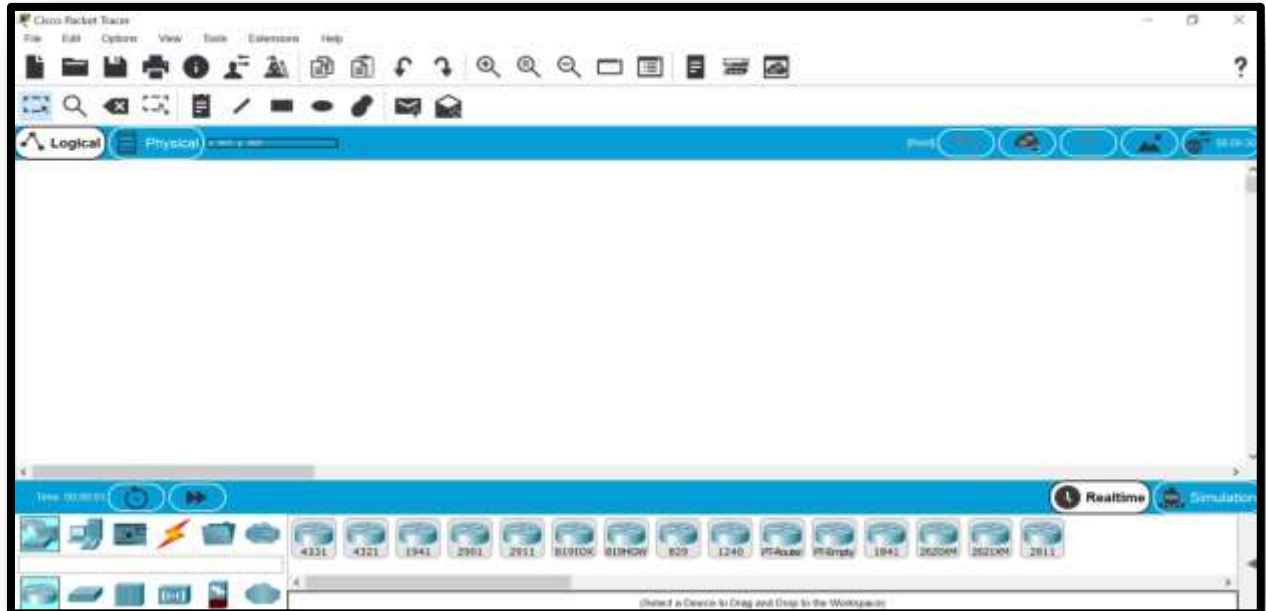
Ethernet straight-through cables (auto-selected)

4. Methodology / Procedure

Step 1: Initial Setup

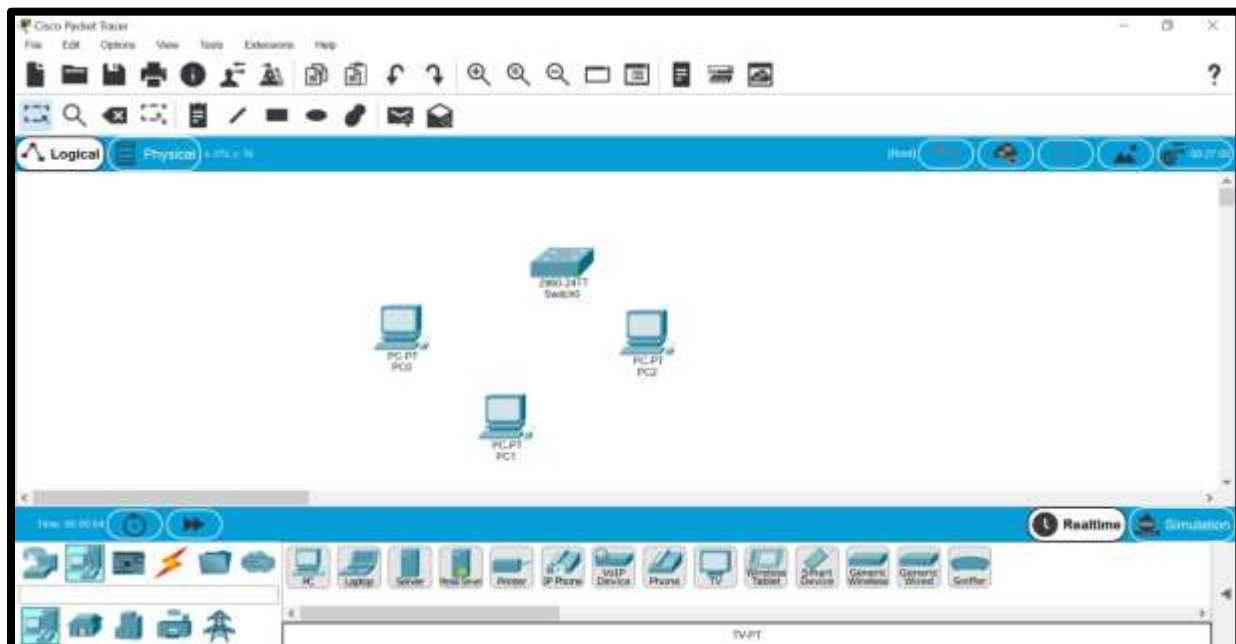
A new project was created in Cisco Packet Tracer, and the workspace was prepared for device placement.

Star Topology Implementation



Step 2: Device Placement

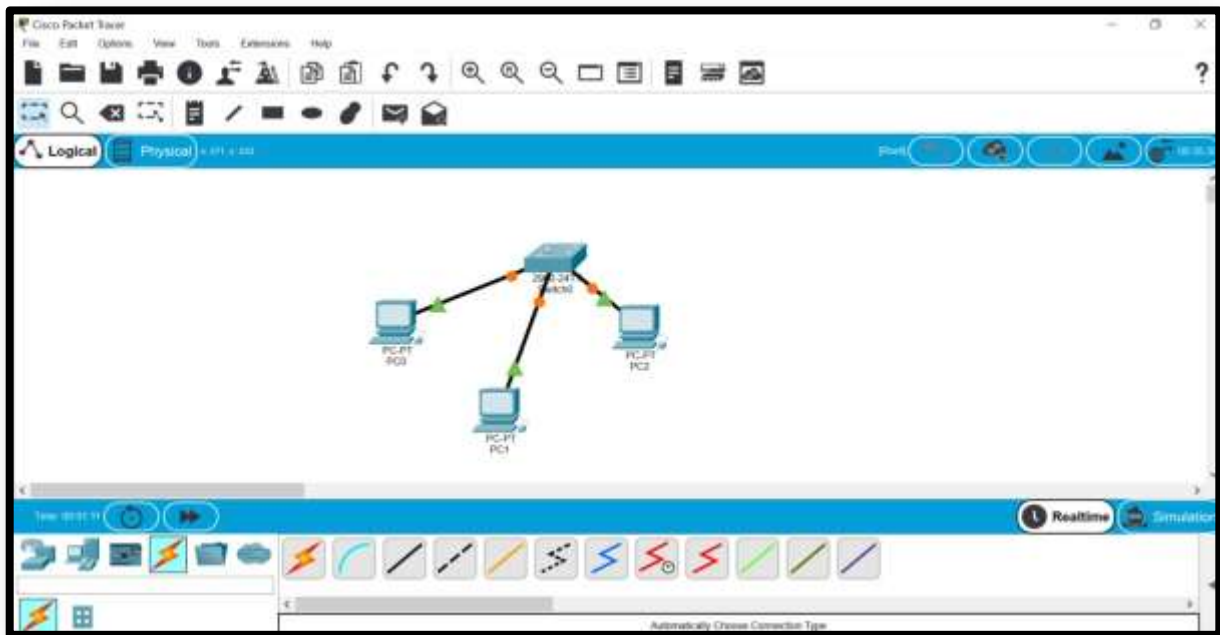
A 2960-24TT switch was added to the workspace. Three PCs were then inserted and labeled as PC0, PC1, and PC2.



Star Topology Implementation

Step 3: Establishing Connections

Each PC was connected to the switch using straight-through Ethernet cables, forming a star topology. Automatic cable selection was used for convenience.



Step 4: IP Configuration

IP addresses were assigned through the Desktop → IP Configuration section:

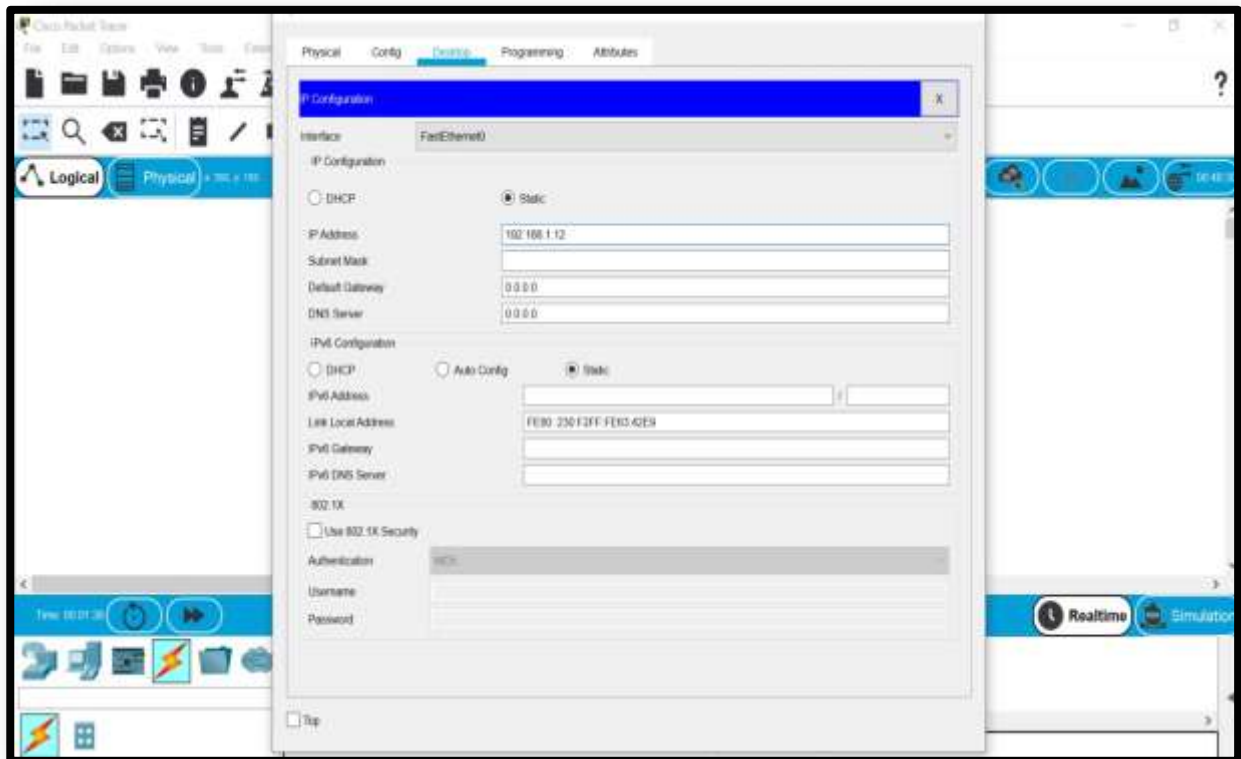
PC0: 192.168.1.12

PC1: 192.168.1.13

PC2: 192.168.1.14

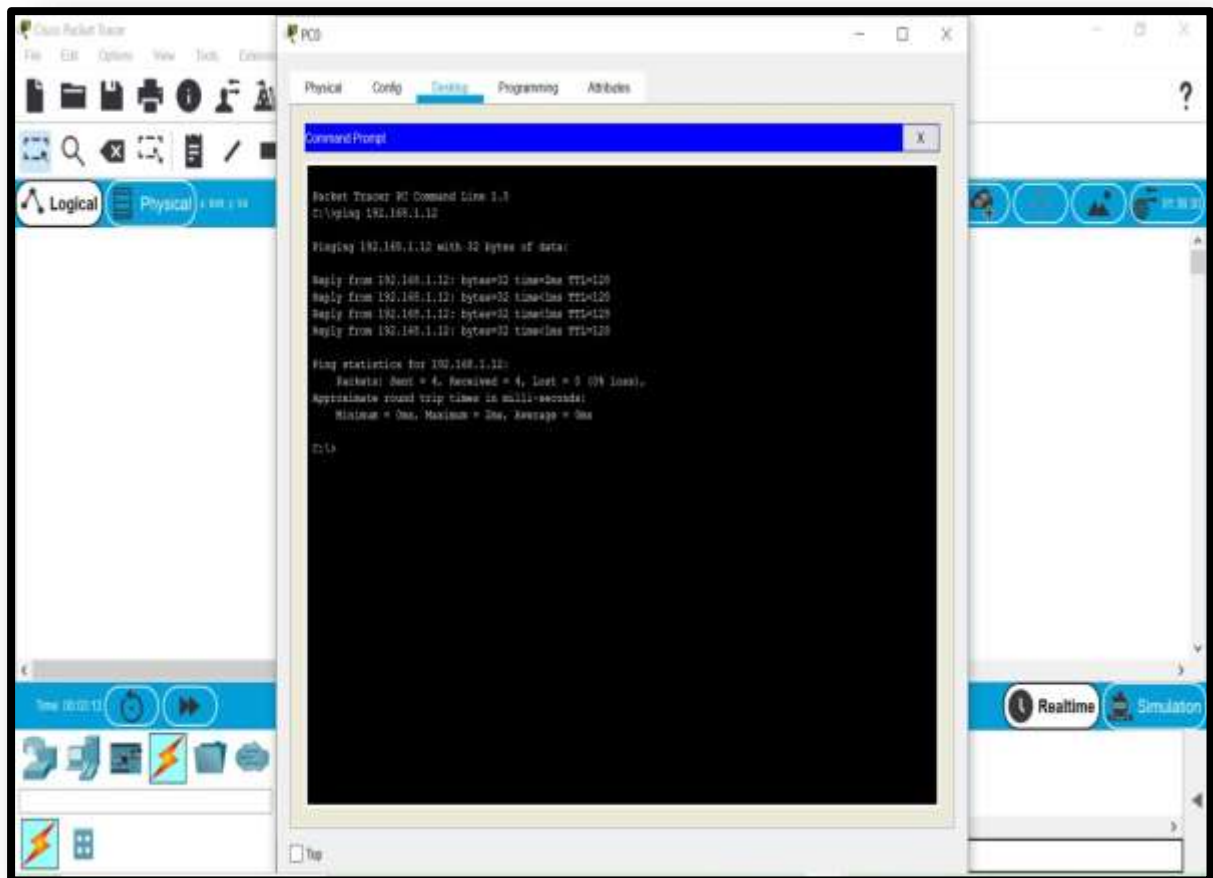
All devices were configured within the same subnet.

Star Topology Implementation



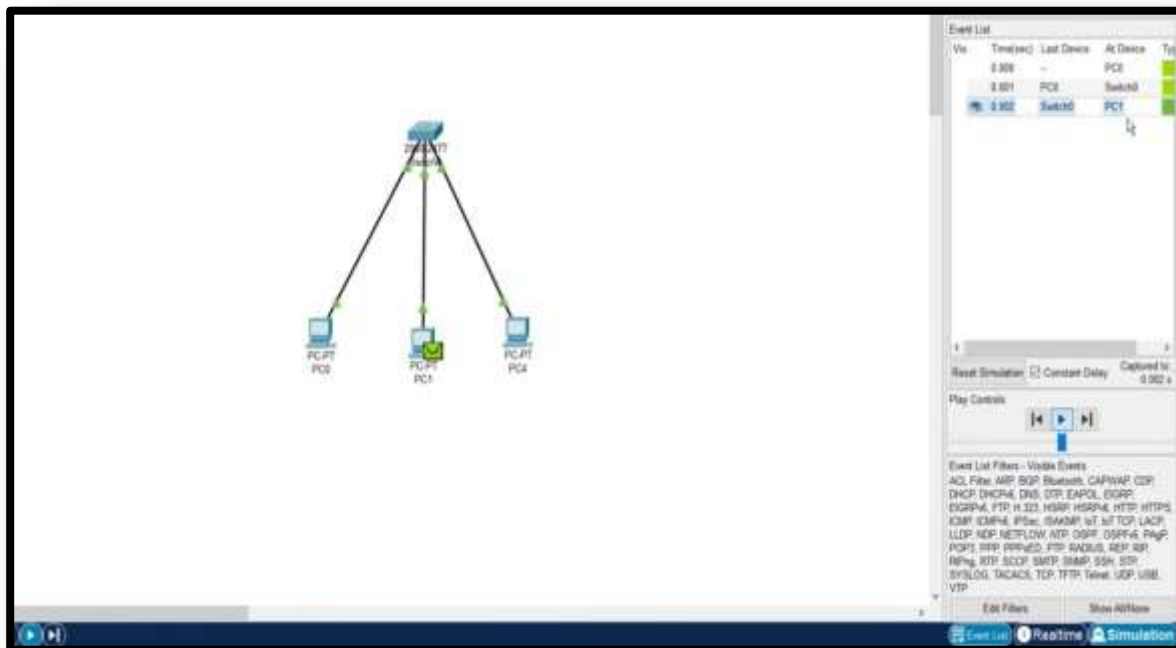
Step 5: Connectivity Testing

The Command Prompt on PC0 was used to send ping requests to other PCs. Successful replies with no packet loss confirmed proper network configuration.



Step 6: Packet Analysis in Simulation Mode

The simulation mode was enabled to observe packet flow. ARP requests and ICMP echo packets were tracked as they moved between devices. The Event List provided detailed information such as source, destination, and protocol type.



5. Results and Observations

The star topology was implemented successfully without any connection issues

All devices communicated effectively with one another

Packet flow, including ARP and ICMP, was clearly visible in Simulation Mode

The switch forwarded frames efficiently to the correct destination

Ping results showed minimal delay (0–2 ms) and zero packet loss

6. Analysis and Discussion

This experiment highlighted the effectiveness of a star topology in managing network communication. The switch played a key role by learning MAC addresses and reducing unnecessary traffic.

Using Simulation Mode provided a deeper understanding of how data travels through the network. It also reinforced concepts related to the OSI model, particularly the Data Link Layer (switch operations) and Network Layer (IP addressing and communication).

7. Conclusion

Star Topology Implementation

The lab objectives were successfully achieved. A functional star topology network was designed, configured, and tested using Cisco Packet Tracer. Observing packet transmission in simulation mode enhanced comprehension of networking concepts and device behavior. This exercise proved valuable in building practical skills related to network configuration, analysis, and troubleshooting.
